Numerical control machine was invented by an American John T. Parsons in 1940. He worked in Massachusetts institute of technology.

9.1 Introduction

Production is a process of converting raw material into finished products. The process of productions is achieved by the collective efforts of man and machine using materials and tools. It requires information and energy to accomplish production by machining. Over the period of time, several techniques and methods are used in the process. The evolution of new methods of production saw the increasing involvement of machines in providing the information and energy required for production and the role played by the humans declined.

Prepared programs consisting of information and instructions took the role of controlling the machines instead of manual control. This has led to a manufacturing system of higher production at lower cost with more accuracy.



CNC Machine

9.2 Numerical control

In NC system, intervention of human beings in the machining process is substituted by some operating instructions in a coded form. This coding, otherwise known as part program, is stored in cards or tapes. Getting the required instructions from these input media, the machine carries out different tasks in a proper sequence.

Numerical control can be defined as a system in which actions of a machine tool are controlled by recorded informations in the form of numerical data.

9.3 Computer Numerical Control

When the activities of a numerical control machine is administered by a computer, it is known as computer numerical control machine tool. The functions of the machine tool are controlled by the instructions stored as programs in the computer.

9.4 Elements of CNC machines

As explained earlier, the NC system requires the preparation of part program based on the product drawing, preparation of input media (punched cards & punched tapes), the data entry into the control unit, consequent processing and actuation of the machine tool to produce the desired part.

To perform all the above operations the NC machine have the following elements

1. Software
2. Machine control unit
3. Driving devices
4. Machine tool

XII BME_English version CHAPTER 09.indd 156 1/8/2020 1:38:49 PM